

This document provides additional assistance with wiring your Extron IP Link Pro Control Processor to your device. Different components may require a different wiring scheme than those listed below.

For complete operating instructions, refer to the user's manual for the specific IP Link Pro Control Processor or the documentation supplied by the manufacturer of the controlled device.

For more information on using Global Scriptor Modules, refer to the "[Guide to Using Scriptor Modules](#)" document.

Device Specifications

Device Type: Video Projector
Manufacturer: Casio
Firmware Version: N/A
Model(s): XJ-S400U, XJ-S400W, XJ-S400UN, XJ-S400WN

Tested on the Following Software and Firmware Versions

IP Link Pro Control Processor Firmware	Global Scriptor Version
3.03.0000-b006	2.3.0

Version History

Module Version	Date	Notes
1_1_0_0	5/22/2019	Initial Version

Module Notes

- Unidirectional variable must be set to 'True' if status is not required. Default value is 'False'.
Example: `InterfaceName.Unidirectional = 'True'`
- connectionCounter variable must be set to the number of queries that will be sent to the device before displaying 'Disconnected' if no response is received. Default value is 15.
Example: `InterfaceName.connectionCounter = 5`
- If login credentials are required, devicePassword must be set accordingly. deviceUsername default value is adminrespectively.
Example: `InterfaceName.devicePassword = 'extron'`
- Ethernet control is only available for models: XJ-S400UN and XJ-S400WN.

Supported Classes and Examples

SerialClass
<code>InterfaceName = ModuleName.SerialClass(ProcessorName, 'COM1', Model='XJ-S400U')</code>
SerialOverEthernetClass
<code>InterfaceName = ModuleName.SerialOverEthernetClass('192.168.254.254', 2001, Model='XJ-S400U')</code>
EthernetClass
<code>InterfaceName = ModuleName.EthernetClass('192.168.254.254', 4352, Model='XJ-S400U')</code>

Control Commands (Serial)

Format with Qualifier:

```
InterfaceName.Set(Command, Value, {'Qualifier Key': 'Qualifier Value'})
```

Format with Qualifier:

```
InterfaceName.Set(Command, Value)
```

Command AspectRatio ³	Value 'Normal (RGB)' 'LetterBox' '4:3'	Value '16:9' 'Full' '16:10'	Value 'Normal (Video)' 'True'
# AspectRatio example InterfaceName.Set('AspectRatio', 'Normal (RGB)')			
Command AudioMute	Value 'On'	Value 'Off'	
# AudioMute example InterfaceName.Set('AudioMute', 'On')			
Command Freeze	Value 'On'	Value 'Off'	
# Freeze example InterfaceName.Set('Freeze', 'On')			
Command Input	Value 'RGB 1' ² 'Auto 1' 'Template' 'Network' 'RGB' ¹	Value 'Component 1' ² 'HDMI 1' 'RGB 2' ² 'Auto 2' ² 'Component' ¹	Value 'Video' 'HDMI 2' 'Component 2' ² 'Casio USB Tool' ² 'Auto' ¹
# Input example InterfaceName.Set('Input', 'RGB 1')			
Command LightControl	Value 'Bright' 'Light Output 7' 'Light Output 4' 'Light Output 1'	Value 'Normal' 'Light Output 6' 'Light Output 3'	Value 'Light Sensor On' 'Light Output 5' 'Light Output 2'
# LightControl example InterfaceName.Set('LightControl', 'Bright')			
Command MenuNavigation	Value 'Up' 'Right' 'Menu'	Value 'Down' 'Enter'	Value 'Left' 'Escape'
# MenuNavigation example InterfaceName.Set('MenuNavigation', 'Up')			
Command Power	Value 'On'	Value 'Off'	
# Power example InterfaceName.Set('Power', 'On')			
Command VideoMute	Value 'On'	Value 'Off'	
# VideoMute example InterfaceName.Set('VideoMute', 'On')			
Command Volume	Value 0 to 30 in steps of 1		

```
# Volume example  
InterfaceName.Set('Volume', 30)
```

¹ Only available with models XJ-S400U and XJ-S400W.

² Only available with models XJ-S400UN and XJ-S400WN.

³ Availability of values depends on the Input type.

Status Available (Serial)

For all commands, call Update to receive the latest status. ConnectionStatus does not support the Update function and is triggered by the device providing a successful response to other Update function calls.

Format with Qualifier:

```
InterfaceName.Update(Command, {'Qualifier Key': 'Qualifier Value'})
Value = InterfaceName.ReadStatus(Command, {'Qualifier Key': 'Qualifier Value'})
InterfaceName.SubscribeStatus(Command, {'Qualifier Key': 'Qualifier Value'},
```

FeedbackHandler)

FeedbackHandler will be called only when the specified qualifier gets a new status.

Format without Qualifier:

```
InterfaceName.Update(Command)
Value = InterfaceName.ReadStatus(Command)
InterfaceName.SubscribeStatus(Command, None, FeedbackHandler)
FeedbackHandler will be called when any qualifier gets a new status.
```

Command AspectRatio ³	Value 'Normal (RGB)' 'LetterBox' '4:3'	Value '16:9' 'Full' '16:10'	Value 'Normal (Video)' 'True'
# AspectRatio example InterfaceName.Update('AspectRatio') Value = InterfaceName.ReadStatus('AspectRatio') InterfaceName.SubscribeStatus('AspectRatio', None, FeedbackHandler)			
Command AudioMute	Value 'On'	Value 'Off'	
# AudioMute example InterfaceName.Update('AudioMute') Value = InterfaceName.ReadStatus('AudioMute') InterfaceName.SubscribeStatus('AudioMute', None, FeedbackHandler)			
Command ConnectionStatus	Value 'Connected'	Value 'Disconnected'	
Value = InterfaceName.ReadStatus('ConnectionStatus') InterfaceName.SubscribeStatus('ConnectionStatus', None, FeedbackHandler)			
Command DeviceStatus	Value 'Normal' 'Light Error'	Value 'Fan Error' 'Other Error'	Value 'Temperature Error'
# DeviceStatus example InterfaceName.Update('DeviceStatus') Value = InterfaceName.ReadStatus('DeviceStatus') InterfaceName.SubscribeStatus('DeviceStatus', None, FeedbackHandler)			
Command Freeze	Value 'On'	Value 'Off'	
# Freeze example InterfaceName.Update('Freeze') Value = InterfaceName.ReadStatus('Freeze') InterfaceName.SubscribeStatus('Freeze', None, FeedbackHandler)			
Command Input	Value 'RGB 1' ² 'Auto 1' 'Template' 'Network' 'RGB' ¹	Value 'Component 1' ² 'HDMI 1' 'RGB 2' ² 'Auto 2' ² 'Component' ¹	Value 'Video' 'HDMI 2' 'Component 2' ² 'Casio USB Tool' ² 'Auto' ¹
# Input example			

<pre>InterfaceName.Update('Input') Value = InterfaceName.ReadStatus('Input') InterfaceName.SubscribeStatus('Input', None, FeedbackHandler)</pre>			
Command	Value		
LampUsage	Hours		
<pre># LampUsage example InterfaceName.Update('LampUsage') Value = InterfaceName.ReadStatus('LampUsage') InterfaceName.SubscribeStatus('LampUsage', None, FeedbackHandler)</pre>			
Command	Value	Value	Value
LightControl	'Bright'	'Normal'	'Light Sensor On'
	'Light Output 7'	'Light Output 6'	'Light Output 5'
	'Light Output 4'	'Light Output 3'	'Light Output 2'
	'Light Output 1'		
<pre># LightControl example InterfaceName.Update('LightControl') Value = InterfaceName.ReadStatus('LightControl') InterfaceName.SubscribeStatus('LightControl', None, FeedbackHandler)</pre>			
Command	Value	Value	
Power	'On'	'Off'	
<pre># Power example InterfaceName.Update('Power') Value = InterfaceName.ReadStatus('Power') InterfaceName.SubscribeStatus('Power', None, FeedbackHandler)</pre>			
Command	Value	Value	
VideoMute	'On'	'Off'	
<pre># VideoMute example InterfaceName.Update('VideoMute') Value = InterfaceName.ReadStatus('VideoMute') InterfaceName.SubscribeStatus('VideoMute', None, FeedbackHandler)</pre>			
Command	Value		
Volume	0 to 30 in steps of 1		
<pre># Volume example InterfaceName.Update('Volume') Value = InterfaceName.ReadStatus('Volume') InterfaceName.SubscribeStatus('Volume', None, FeedbackHandler)</pre>			

¹ Only available with models XJ-S400U and XJ-S400W.² Only available with models XJ-S400UN and XJ-S400WN.³ Availability of values depends on the Input type.

Control Commands (Ethernet)

Format with Qualifier:

```
InterfaceName.Set(Command, Value, {'Qualifier Key': 'Qualifier Value'})
```

Format with Qualifier:

```
InterfaceName.Set(Command, Value)
```

Command	Value	Value	
Freeze	'On'	'Off'	
# Freeze example InterfaceName.Set('Freeze', 'On')			
Command	Value	Value	Value
Input	'RGB 1' ²	'Component 1' ²	'Video'
	'Auto 1'	'HDMI 1'	'HDMI 2'
	'Template'	'RGB 2' ²	'Component 2' ²
	'Network'	'Auto 2' ²	'Casio USB Tool' ²
	'RGB' ¹	'Component' ¹	'Auto' ¹
# Input example InterfaceName.Set('Input', 'RGB 1')			
Command	Value	Value	
Power	'On'	'Off'	
# Power example InterfaceName.Set('Power', 'On')			
Command	Value	Value	
Volume	'Up'	'Down'	
Qualifier Key	Qualifier Value	Qualifier Value	
'Device'	'Speaker'	'Microphone'	
# Volume example InterfaceName.Set('Volume', 'Up', {'Device': 'Speaker'})			

¹ Only available with models XJ-S400U and XJ-S400W.

² Only available with models XJ-S400UN and XJ-S400WN.

Status Available (Ethernet)

For all commands, call Update to receive the latest status. ConnectionStatus does not support the Update function and is triggered by the device providing a successful response to other Update function calls.

Format with Qualifier:

```
InterfaceName.Update(Command, {'Qualifier Key': 'Qualifier Value'})
Value = InterfaceName.ReadStatus(Command, {'Qualifier Key': 'Qualifier Value'})
InterfaceName.SubscribeStatus(Command, {'Qualifier Key': 'Qualifier Value'},
```

FeedbackHandler)

FeedbackHandler will be called only when the specified qualifier gets a new status.

Format without Qualifier:

```
InterfaceName.Update(Command)
Value = InterfaceName.ReadStatus(Command)
InterfaceName.SubscribeStatus(Command, None, FeedbackHandler)
FeedbackHandler will be called when any qualifier gets a new status.
```

Command ConnectionStatus	Value 'Connected'	Value 'Disconnected'	
<pre>Value = InterfaceName.ReadStatus('ConnectionStatus') InterfaceName.SubscribeStatus('ConnectionStatus', None, FeedbackHandler)</pre>			
Command DeviceStatus	Value 'Normal' 'Lamp Warning' 'Temperature Error' 'Multiple Warnings/Errors'	Value 'Fan Warning' 'Lamp Error' 'Other Warning'	Value 'Fan Error' 'Temperature Warning' 'Other Error'
<pre># DeviceStatus example InterfaceName.Update('DeviceStatus') Value = InterfaceName.ReadStatus('DeviceStatus') InterfaceName.SubscribeStatus('DeviceStatus', None, FeedbackHandler)</pre>			
Command FilterUsage	Value Hours		
<pre># FilterUsage example InterfaceName.Update('FilterUsage') Value = InterfaceName.ReadStatus('FilterUsage') InterfaceName.SubscribeStatus('FilterUsage', None, FeedbackHandler)</pre>			
Command Freeze	Value 'On'	Value 'Off'	
<pre># Freeze example InterfaceName.Update('Freeze') Value = InterfaceName.ReadStatus('Freeze') InterfaceName.SubscribeStatus('Freeze', None, FeedbackHandler)</pre>			
Command Input	Value 'RGB 1' ² 'Auto 1' 'Template' 'Network' 'RGB' ¹	Value 'Component 1' ² 'HDMI 1' 'RGB 2' ² 'Auto 2' ² 'Component' ¹	Value 'Video' 'HDMI 2' 'Component 2' ² 'Casio USB Tool' ² 'Auto' ¹
<pre># Input example InterfaceName.Update('Input') Value = InterfaceName.ReadStatus('Input') InterfaceName.SubscribeStatus('Input', None, FeedbackHandler)</pre>			
Command LampUsage	Value Hours		

Global Scripter Module Communication Sheet

<pre># LampUsage example InterfaceName.Update('LampUsage') Value = InterfaceName.ReadStatus('LampUsage') InterfaceName.SubscribeStatus('LampUsage', None, FeedbackHandler)</pre>			
Command	Value	Value	Value
Power	'On'	'Off'	'Cooling Down'
	'Warming Up'		
<pre># Power example InterfaceName.Update('Power') Value = InterfaceName.ReadStatus('Power') InterfaceName.SubscribeStatus('Power', None, FeedbackHandler)</pre>			

¹ Only available with models XJ-S400U and XJ-S400W.

² Only available with models XJ-S400UN and XJ-S400WN.

Cable and Adapter Requirements

Captive Screw to Female DB9 RS-232 Serial Cable

Notes for the Device

Serial communication

Port Type: RS-232

Baud Rate: 19200

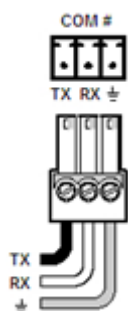
Data Bits: 8

Parity: None

Stop Bits: One

Flow Control: None

Pin Assignments Diagram



Please contact manufacturer for Pin Assignments Diagram



Network communication

When configuring the Ethernet module, be sure device settings match those of the Global Scriptor ethernet interface.

Port Type:	Ethernet
Default Port:	4352
Logon Credentials Supported:	Yes
Default Password:	admin
Multi-Connection Capabilities:	Undetermined
Port Changeability:	Yes

Ethernet Module Configuration Description

Please refer to user manual for settings and changes to the network communication

Notes for the Device

If the projector does not have a password saved or the security function of the projector is turned off, then please leave the password field empty.

Appendix A. Set Commands (Serial)

Aspect Ratio 16:10	(ARZ7)
Aspect Ratio 16:9	(ARZ1)
Aspect Ratio 4:3	(ARZ6)
Aspect Ratio Full	(ARZ4)
Aspect Ratio LetterBox	(ARZ3)
Aspect Ratio Normal (RGB)	(ARZ0)
Aspect Ratio Normal (Video)	(ARZ2)
Aspect Ratio True	(ARZ5)
Audio Mute Off	(MUT0)
Audio Mute On	(MUT1)
Freeze Off	(FRZ0)
Freeze On	(FRZ1)
Input Auto 1	(SRC6)
Input Auto 2	(SRC10)
Input Casio USB Tool	(SRC13)
Input Component 1	(SRC1)
Input Component 2	(SRC4)
Input HDMI 1	(SRC7)
Input HDMI 2	(SRC14)
Input Network	(SRC8)
Input RGB 1	(SRC0)
Input RGB 2	(SRC3)
Input Template	(SRC15)
Input Video	(SRC2)
Light Control Bright	(PMD0)
Light Control Light Output 1	(PMD9)
Light Control Light Output 2	(PMD8)
Light Control Light Output 3	(PMD7)
Light Control Light Output 4	(PMD6)
Light Control Light Output 5	(PMD5)
Light Control Light Output 6	(PMD4)
Light Control Light Output 7	(PMD3)
Light Control Light Sensor On	(PMD2)
Light Control Normal	(PMD1)
Menu Navigation Down	(KEY2)
Menu Navigation Enter	(KEY5)
Menu Navigation Escape	(KEY6)
Menu Navigation Left	(KEY3)
Menu Navigation Menu	(KEY11)
Menu Navigation Right	(KEY4)
Menu Navigation Up	(KEY1)
Power Off	(PWR0)

Global Scriptor Module Communication Sheet

Revision: 5/22/2019

Power On	(PWR1)
Video Mute Off	(BLK0)
Video Mute On	(BLK1)
Volume 0	(VOL0)
Volume 30	(VOL30)
Volume Down Device Microphone	(VOLDown)
Volume Down Device Speaker	(VOLDown)
Volume Up Device Microphone	(VOLUp)
Volume Up Device Speaker	(VOLUp)

Appendix B. Update Commands (Serial)

Aspect Ratio	(ARZ?)
Audio Mute	(MUT?)
Device Status	(STS?)
Freeze	(FRZ?)
Input	(SRC?)
Lamp Usage	(LMP?)
Light Control	(PMD?)
Power	(PWR?)
Video Mute	(BLK?)
Volume	(VOL?)

Appendix C. Set Commands (Ethernet)

Freeze Off	%2FREZ 0\x0D
Freeze On	%2FREZ 1\x0D
Input Auto 1	%1INPT 11\x0D
Input Auto 2	%1INPT 14\x0D
Input Casio USB Tool	%1INPT 43\x0D
Input Component 1	%1INPT 13\x0D
Input Component 2	%1INPT 16\x0D
Input HDMI 1	%1INPT 31\x0D
Input HDMI 2	%1INPT 32\x0D
Input Network	%1INPT 44\x0D
Input RGB 1	%1INPT 12\x0D
Input RGB 2	%1INPT 15\x0D
Input Template	%1INPT 45\x0D
Input Video	%1INPT 21\x0D
Power Off	%1POWR 0\x0D
Power On	%1POWR 1\x0D
Volume Down Device Microphone	%2MVOL 0\x0D
Volume Down Device Speaker	%2SVOL 0\x0D
Volume Up Device Microphone	%2MVOL 1\x0D
Volume Up Device Speaker	%2SVOL 1\x0D

Appendix D. Update Commands (Ethernet)

Device Status	%1ERST ?\x0D
Filter Usage	%2FILT ?\x0D
Freeze	%2FREZ ?\x0D
Input	%1INPT ?\x0D
Lamp Usage	%1LAMP ?\x0D
Power	%1POWR ?\x0D